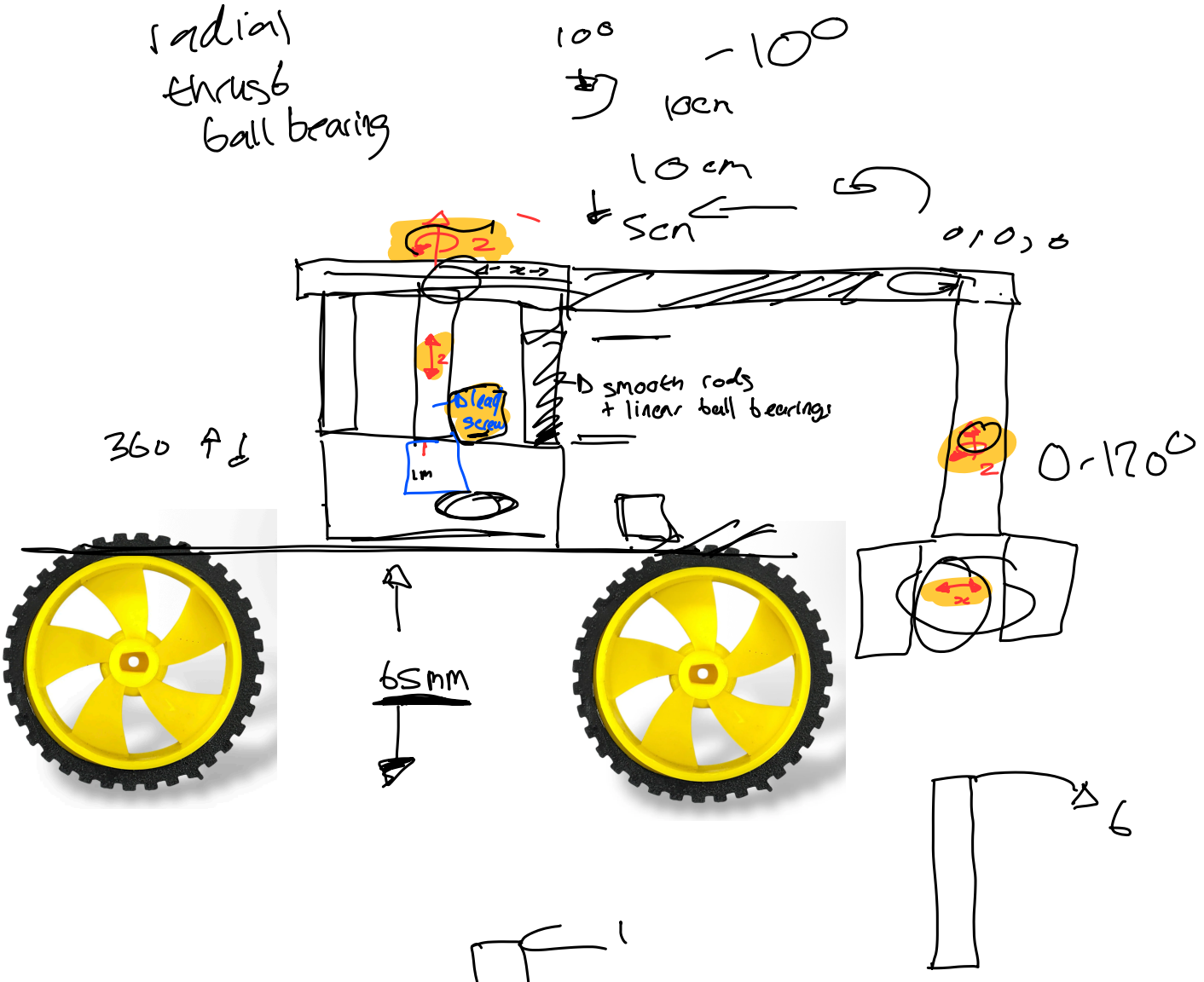
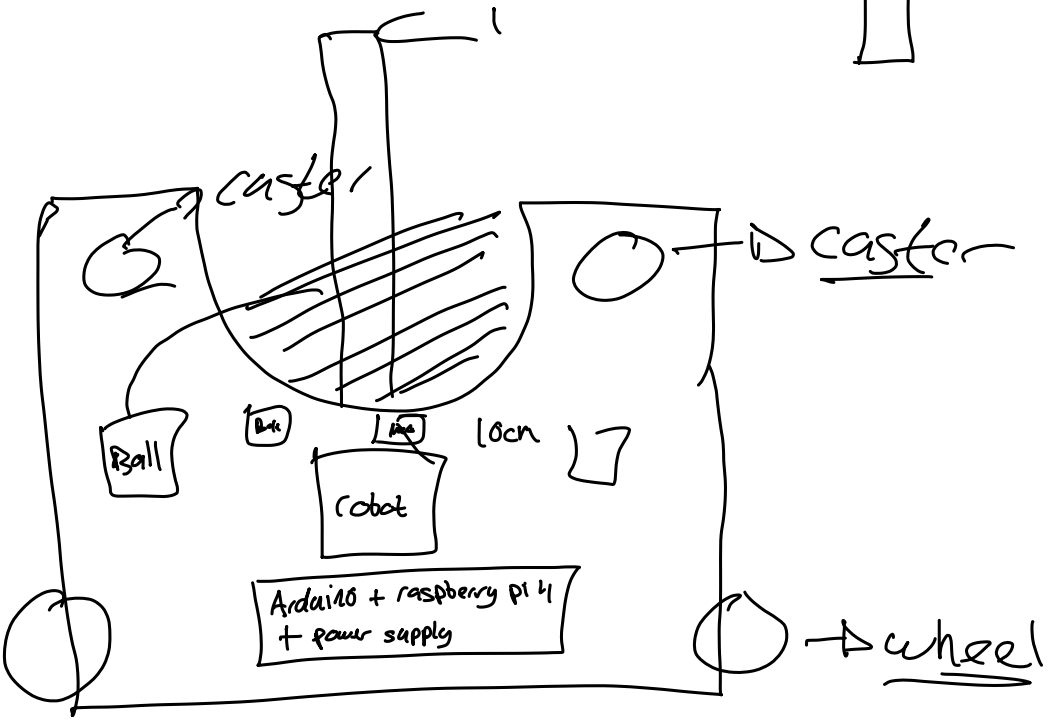
radial
thrust
ball bearing

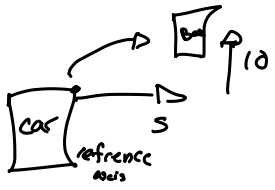


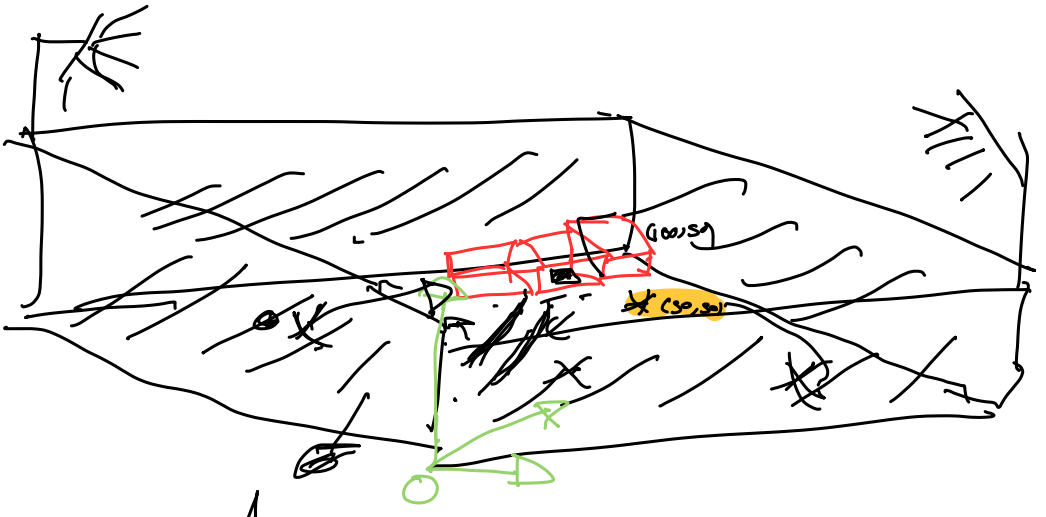


start → object → Box
Calculations calculations

Start → object → Box
calcu

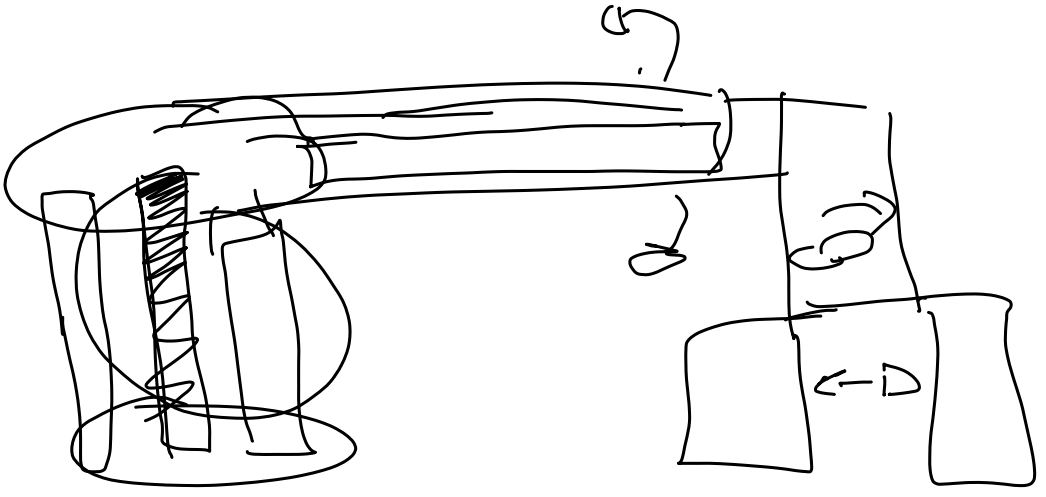
100 x 180 cm





Acrylic

1. cameras scan for objects (V→B→E→C→B)
2. Scan for boxes



Joint 1: Base joint (0-360°), rotation

- Stepper motor + maybe small gear system
- Bearings
- Micro limit switch

Joint 2: Vertical, linear

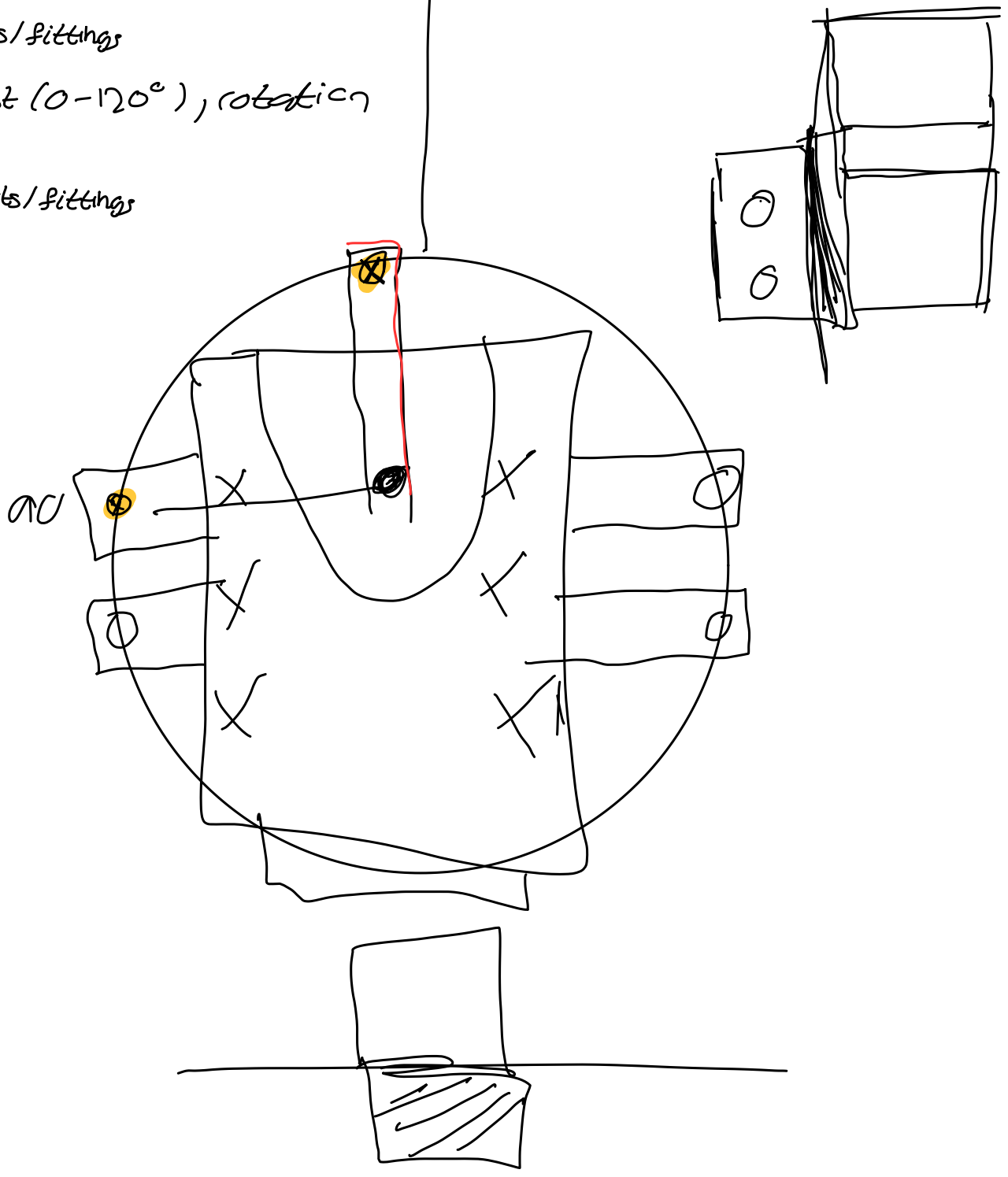
- Stepper motor
- Lead screw
- smooth rods
- linear ball bearing
- ball bearings
- Shaft coupler
- micro limit switch

Joint 3: Elbow (0-120°), rotation

- Servo
- Connecting parts/fittings

Join 4: Wrist (0-120°), rotation

- Servo
- Connecting parts/fittings

Platform:

- Base of the robot
- Boxes coming out
- Arduino R3
- Raspberry pi 4
- Battery 12V
- Breadboard
- Motor drive (HG488/PCA9685)
- Step down converter
- Camera (subject to change)